Solving Discrete Time Heterogeneous Agent Models with Aggregate Risk and Many Idiosyncratic States by Perturbation (by Bayer, C. and Luetticke, R.)

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November 30, 2022

Introduction

- HA models with aggregate uncertainty have become widely used in macroeconomics.
- The curse of dimensionality ($\because$ many idiosyncratic shocks) $\Rightarrow$ the dimensionality reduction is required.
- This paper contributes to the literature by providing an efficient algorithm that reduces dimensionality while keeping information of the stationary equilibrium.

Contributions

- Reduce dimensionality before solving the stationary equilibrium and then linearize (*Bungartz and Griebel 2004; Krueger and Kubler 2004; Reiter 2009; Winberry 2018*).
  - **Problem**: Impose a numerical constraint on the stationary equilibrium.
- Reduce dimensionality after linearization (*Reiter 2010; Ahn et al. 2017*).
  - **Problem**: Calculation of a large Jacobian is required.
- This paper proposes a dimensionality reduction **after** solving the stationary equilibrium but **before** perturbing the system.

Prerequisites and Notation

- The aggregate states $S_t \in \mathbb{R}^n$ and the idiosyncratic states $s_{it} \in \mathbb{R}^m$:

$$S_t = \begin{bmatrix} X_t \\ D_t \end{bmatrix}, \quad s_{it} = \begin{bmatrix} x_{it} \\ d_{it} \end{bmatrix} \tag{1}$$

  - X_t and x_{it} are exogenous stochastic components.
  - D_t and d_{it} are endogenous deterministic components.

- Law of motions for stochastic variables and aggregate states:

$$X_{t+1} = H^X(X_t) + \varepsilon_{t+1}, \quad Var(\varepsilon_{t+1}) = \omega\Omega \tag{2}$$

$$x_{it+1} = h^x(x_{it}) + \epsilon_{it+1}, \quad Var(\epsilon_{it+1}) = \sigma\Sigma \tag{3}$$

$$D_{t+1} = H^D(X_t, D_t, \mu_t) \tag{4}$$

Prerequisites and Notation

- The household's problem:

$$v(x_{it}, d_{it}, S_t, \mu_t) = \max_{d_{it+1}} u(x_{it}, d_{it}, d_{it+1}; P_t) + \beta \mathbb{E} v(x_{it+1}, d_{it+1}, S_{t+1}, \mu_{t+1}) \tag{5}$$

$$s.t.\ d_{it+1} \in \Gamma(x_{it}, d_{it}, P_t). \tag{6}$$

  - μ_t is the *distribution function* of agents over s_{it}.
  - $P_t = P(X_t, D_t, \mu_t)$ is a pricing kernel.

A Stationary Equilibrium: No Aggregate Risk

- A stationary equilibrium corresponds to the equilibrium without aggregate risk, that is, where $\omega = 0$.
- How to solve for a stationary equilibrium?
  - Discretize the state space and use the value function iteration.
  - Optimal policy $\bar{h}_t(s_{it}; P)$ induces flow utility $\bar{u}_{\bar{h}}$ and transition probability matrix $\Pi_{\bar{h}}$.
- A stationary equilibrium using a full tensor grid can be characterized by the following equations:

$$d\bar{\mu} = d\bar{\mu}\Pi_{\bar{h}}. \tag{7}$$

$$\bar{v} = u_{\bar{h}^d} + \beta \Pi_{\bar{h}} \bar{v}. \tag{8}$$

A Sequential Equilibrium: Introducing Aggregate Risk

- Uncertainty in aggregate states:
  - Value function written as functions of idiosyncratic states are no longer time constant in equilibrium.
  - Prices and distributions change over time.

- Sequential equilibrium conditions:

$$\mathbb{E}_t F(d\mu_t, S_t, d\mu_{t+1}, v_t, P_t, v_{t+1}, P_{t+1}, \epsilon_{t+1}) = \mathbb{E}_t \begin{bmatrix} d\mu_{t+1} - d\mu_t \Pi_{h_t} \\ X_{t+1} - H^X(X_t, D_t) + \epsilon_{t+1} \\ D_{t+1} - H^D(X_t, D_t, d\mu_t) \\ v_t - (u_{h_t^d} + \beta \Pi_{h_t} v_{t+1}) \\ \Phi_t(h_t^d, d\mu_t) \\ \epsilon_{t+1} \end{bmatrix} = 0. \tag{9}$$

Approximations

- How to solve the nonliner difference equation $\mathbb{E}_t F = 0$?
- First-order perturbation $\Rightarrow$ calculate the Jacobian of the system and solve the linearized difference equation by relating its solution to the generalized eigenvalue problem.
- Problem? The curse of dimensionality $\Rightarrow$ Reducing dimensionality is required.
  - 7 income grid points $\times$ 100 illiquid asset states $\times$ 100 liquid asset states $\Rightarrow$ more than 70,000 controls in F.

Dimensionality Reduction

- The node values of the value functions:

$$\hat{v}_t(s) = g_v(s; \theta_t, \bar{v}). \tag{10}$$

- How to choose g_v? Use the *discrete cosine transformation* (DCT) of the stationary equilibrium value function.

Dimensionality Reduction

- $\bar{\Theta} = dct(\bar{v})$: the coefficients obtained from DCT of the value function in the stationary equilibrium.
- $\mathcal{I}$: an index set that contains the x percent largest elements from $\bar{\Theta}$.
- θ_t: a sparse vector with non-zero entries only for elements $i \in \mathcal{I}$.
- Define:

$$\hat{\Theta}(\theta_t) = \begin{cases} \bar{\Theta} + \theta_t(i) & i \in \mathcal{I} \\ \bar{\Theta}, & \text{otherwise} \end{cases} \tag{11}$$

- Reconstruct $v_t = v(\theta_t) = idct(\hat{\Theta}(\theta_t))$. $\Rightarrow$ For $v(0) = \bar{v}$.
- Use $\bar{\Theta}$ as a **reference frame**.

Linearization

- Consider continuous random variables $X_1, \cdots, X_n$ with marginal CDFs $F_i(x) = \Pr[X_i \leq x]$.
- The probability integral transform implies:

$$(U_1, \cdots, U_n) = (F(X_1), \cdots, F(X_d)) \tag{12}$$

  has marginals that are uniformly distributed on the interval $[0, 1]$.
- The **copula** of $(X_1, \cdots, X_n)$ is the joint cumulative distribution function of $(U_1, \cdots, U_n)$:

$$C(u_1, \cdots, u_n) = \Pr[U_1 \leq u_1, \cdots, U_n \leq u_n]\,. \tag{13}$$

Linearization

- Use the fixed Copula to calculate an approximate joint distribution from marginal distributions $\mu_t(s) \equiv \bar{C}\{\mu_{1t}(s), \mu_{2t}(s), \cdots, \mu_{mt}(s)\}$.
  - Some moments of the distribution do not matter for aggregate dynamics (*Krusell and Smith 1998*)
- Under this approach, the dynamic system F replaces value functions and distributions by parameters $\theta_t, d_{\mu_{1t}}, \cdots, d_{\mu_{mt}}$, where the $d\mu-$ terms are the histograms of the marginal distributions.
- The dynamic system F has more equations than unknowns $\Rightarrow$ Use only difference in marginals and the difference on $\mathcal{I}$.

A Simple Krusell-Smith Economy

- **Incomplete markets and TFP**
  - Household productivity can be high or low
  - No contingent claims
  - Households save in capital goods (which they rent out)
  - Households supply labour (disutility) and consume (utility)
  - Aggregate productivity (TFP) follows a log AR(1) process
  - Cobb-Douglas production function

Numerical Performance: Setup

- Solve HH problem with 100 grid points for idiosyncratic capital (total grid size $= 200$).
- Use endogenous grid method.

Numerical Performance: Using DCT for dimensionality reduction

- Reduce dimensionality by using DCT of the stationary equilibrium policy function.
- Compare the approximation for different levels of state-space reductions for the policy functions (i.e. different number of coefficients retained)

Numerical Performance: Using DCT for dimensionality reduction

- Consumption policy function

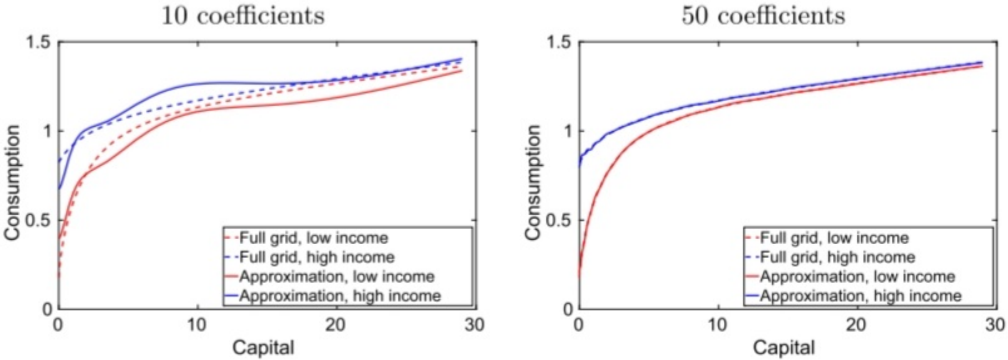


FIGURE 2. Stationary equilibrium consumption policies by sparseness of θ.

Numerical Performance: Using DCT for dimensionality reduction

- Individual policy response to a 20% TFP shock

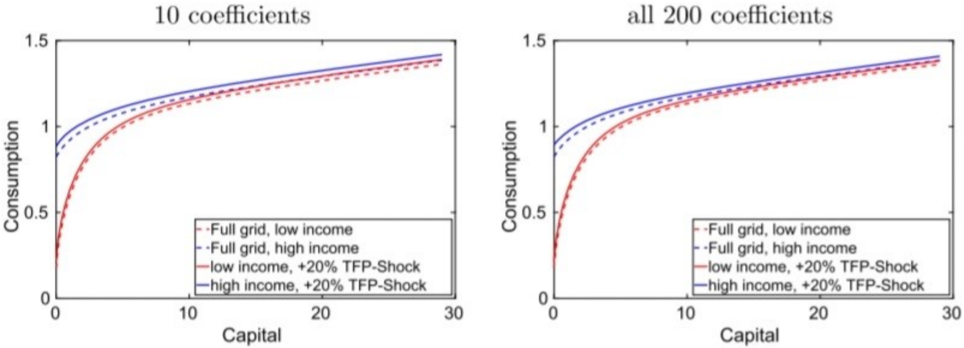


FIGURE 3. Change in consumption policies after a 20% TFP shock by sparseness of θ.

Numerical Performance: Using DCT for dimensionality reduction

- Aggregate response to a 20% TFP shock

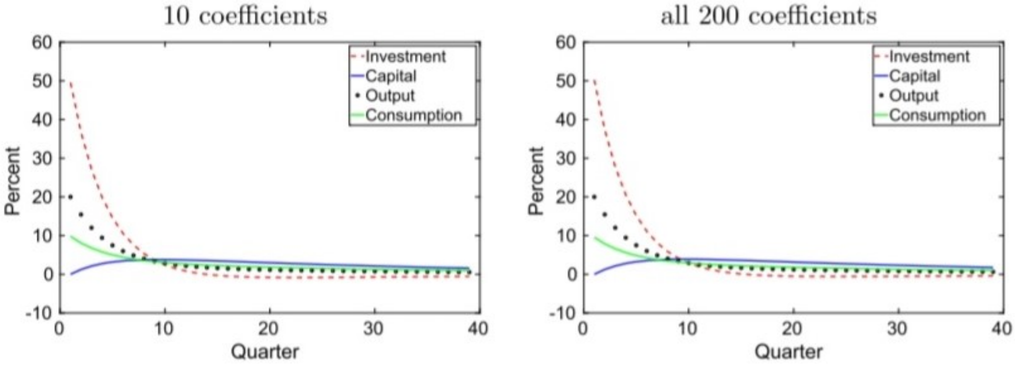


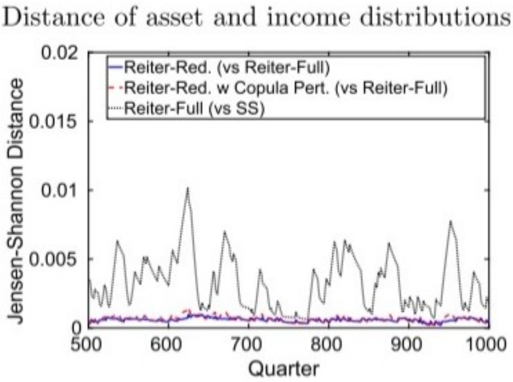
FIGURE 4. Aggregate response after a 20% TFP shock by sparseness of θ.

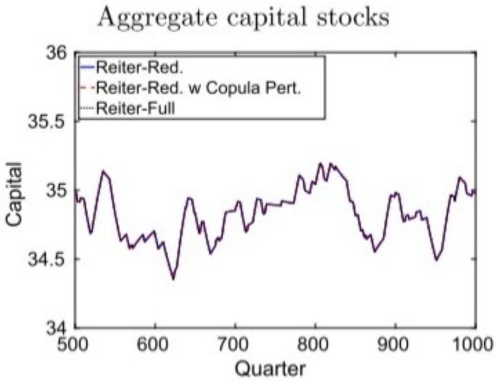
Numerical Performance: Using copula for dimensionality reduction

- Reduce the dimensionality of the distribution function- split it into a copula and marginal distributions.
- Assumption- Copula is time fixed.
- Compare the model implied distributions over time for-
  - **Fixed copula**
  - Use only a few coefficients of the approximation of the copula obtained through a DCT
  - **No reduction in dimensionality**

Numerical Performance: Using copula for dimensionality reduction

- Figures show distance between the distributions with and without the fixed copula assumption (for simulation using TFP shocks)

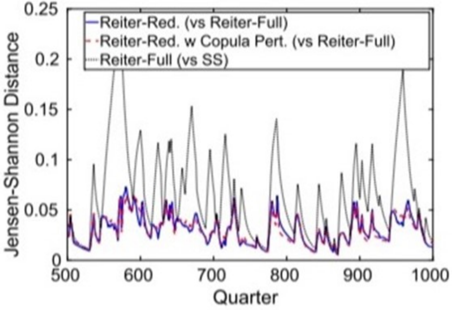




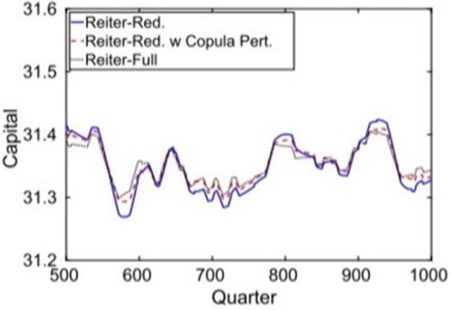
Numerical Performance: Using copula for dimensionality reduction

- Figures show distance between the distributions with and without the fixed copula assumption (for simulation using idiosyncratic income shocks)

Distance of asset and income distributions

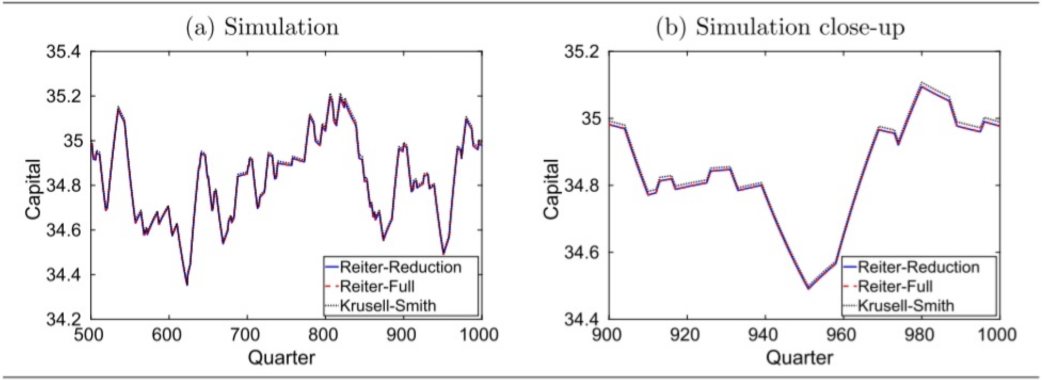


Aggregate capital stocks



Numerical Performance: Simulation Results

- Model simulations using three different solution methods (response of capital to TFP shocks)



Numerical Performance: Error Statistics

- Den Haan error metrics to evaluate accuracy of the solution method

Table 2. Den Haan errors.

	Absolute error (in %) for log capital K_t		
	Reiter-Reduction	Reiter-Full	K-S
Mean	0.0100	0.0102	0.0051
Max	0.0191	0.0193	0.0131

Note: Differences in percent between the simulation of the linearized solutions of the model and simulations in which we solve for the intratemporal equilibrium prices in every period and track the full histogram over time for $t = \{1, \ldots, 1000\}$; see Den Haan (2010a).

Numerical Performance: Computing Time

TABLE 3. Run time for Krusell and Smith model.

	Stationary equilibrium	Krusell and Smith	Reiter-Reduction	Reiter-Full
in seconds	7.05	91.61	0.38	1.19

Note: Run time in seconds on a Dell laptop with an Intel i7-7500U CPU at 2.70 GHz at 4. Model calibration and number of grid points as in Den Haan, Judd, and Juillard (2010). Code in Matlab.

Another Example: A two asset economy with nominal frictions

- Key advantage of linearization with state and control space reduction lies in solving models with many idiosyncratic states.
- Example: K-S model plus
  - a trading cost for capital
  - a perfectly liquid bond
  - a price setting equation for firms
  - a central bank and fiscal authority

Another Example: A two asset economy with nominal frictions

- Analyze an increase in variance of the income shock
  - Requires both assets (100 grid points each)
  - Use 4 income states
  - Use both value functions and consumption policies as controls
  - The full set has $> 120{,}000$ variables
  - This reduces to 204 distribution states and 635 controls for value functions and policies.

Conclusion

- Paper- Proposes to solve heterogeneous agent models with aggregate risk by perturbation.
- Method- Reduce the state space after solving for the stationary equilibrium but before linearizing the non-linear system that characterizes equilibrium dynamics.
- Reduction in state space is achieved by:
  - DCT of the stationary equilibrium value/policy functions.
  - Approximating dynamics of the multidimensional distribution of individual characteristics by a distribution with a fixed copula and varying marginals.
- Increases efficiency and precision of the solution for the equilibrium dynamics of the heterogeneous agent economies.

Discussion and Applications

- How to determine the optimal degree of sparseness?
- According to this method, the coefficients that are retained are those that are important in the stationary equilibrium. It is possible to have coefficients that are not important in the stationary equilibrium but are important outside the stationary equilibrium. This could potentially affect the model's approximations.

Back-up Slides

Definition (A stationary equilibrium without aggregate risk)

A *stationary equilibrium* (without aggregate risk) is a value function $\bar{v}$, a distribution function $\bar{\mu}$, a policy function $\bar{h}^d(s)$, and prices $\bar{P}$ such that:

1. The individual policy function $\bar{h}^d(s)$ is the maximizer of the Bellman equation (4) given $\bar{P}$,
2. The value function solves the Bellman equation (4) given the individual policy function $\bar{h}^d(s)$,
3. Markets clear, that is, $\Phi_t(\bar{h}^d(s), \bar{\mu}) = 0$, and
4. The distribution $\bar{\mu}$ is the stationary distribution of the Markov chain induced by $\bar{h}(s, \epsilon) := \begin{bmatrix} h^x(s) + \epsilon \\ \bar{h}^d(s) \end{bmatrix}$.

Back-up Slides

- Stationary equilibrium:

$$d\bar{\mu} = d\bar{\mu}\Pi_{\bar{h}}. \tag{14}$$

$$\bar{v} = u_{\bar{h}^d} + \beta\Pi_{\bar{h}}\bar{v}. \tag{15}$$

- Solving for the stationary equilibrium is straightforward (more explanation is needed).

Sequential Equilibrium with Recursive Individual Planning

Definition (A sequential competitive equilibrium with recursive individual planning)

A *sequential competitive equilibrium with recursive individual planning* is a sequence of $\left\{v_t, \mu_t, h_t^d(s), S_t, P_t\right\}$ such that at each point in time t:

1. The individual policy $h_t^d(s)$ is the maximizer of the Bellman equation (4) given the prices P_t,
2. The value function solves the Bellman equation (4) given the individual policy $h_t^d(s)$ and the expected continuation value v_{t+1},
3. Market clears, that is, $\Phi(h_t^d, \mu_t, P_t, s_t) = 0$,
4. The distribution μ_{t+1} is induced by $h_t(s, \epsilon) := \begin{bmatrix} h^x(s) + \epsilon_t \\ h_t^d(s) \end{bmatrix}$ and the distribution μ_t, and
5. The sequence of aggregate states is induced by $\begin{bmatrix} X_{t+1} \\ D_{t+1} \end{bmatrix} = \begin{bmatrix} H^X(X_t, D_t) + \epsilon_{t+1} \\ H^D(X_t, D_t, \mu_t) \end{bmatrix}$